WITH
that enable it to do so

MISSILES AND RADARS

- BrahMos hitting missiles with extended range; Advanced Short Range Air-to-Air Missile (ASRAAM), integrated with Jaguar; indigenous Beyond Visual Range (BVR) missile Astra for Su-30, to be inducted soon
- Ground attack missiles like AS-30 and Kh59 for air dominance; long-range air-to-air Python 5 & MICA missiles; SPICE 2000 air-to-ground missiles; HAMMER air-to-ground precision-guided weapon systems
- S-400 air defence system



- The IAF has put in place an Integrated Air Command and Control System (IACCS) that monitors all intelligence/ reconnaissance input across domains
- The various radars in service with the IAF include: Indra air surveillance radar, Rajendra multifunction radar, Swathi weapon-locating mobile radar, tethered aerostat radar; Swordfish long-range tracking radar to counter missiles

However, the sobering fact remains that the current modernisation of the IAF is still a work in progress, and the force is under-strength in frontline fighter aircraft, operating 31 squadrons against a requirement of 42. The pace of acquiring new aircraft has also been painfully slow for decades. The Rafales were India's first major fighter acquisition in 23 years, since the Sukhoi jets were imported from Russia. Even as the country prides itself on its Rafales and Apaches, ancient Mig-21s continue to crash—the latest was in May—and kill pilots. Crashes, often fatal, also bedevil the indigenous Advanced Light Helicopters, Cheetahs and Chetak helicopters operated by the Navy, Army, Coast Guard and the IAF. Thankfully, Mig-21s are already on their way out, and the army will phase out Cheetahs/ Chetaks from 2027. And some of the ambitious indigenous programmes, like the Light Combat Aircraft (LCA)-Mk 2, and the fifth generation stealth Advanced Medium Combat Aircraft (AMCA) are still in the development stage.

"Owing to the volatile and uncertain geopolitical landscape in our region, the need to have a strong and credible air force has become an imperative," IAF chief Air Chief Marshal Vivek Ram Chaudhari said at a recent event. He said the IAF will acquire better tactical abilities where it cannot match the numerical superiority of China's air force. Chaudhari added that the IAF has plans to induct fighter jets, helicopters, missiles and radars worth Rs 2.5-3 lakh crore over the next seven to eight years to tackle the China-Pakistan collusive threat. Chaudhari's comments come at a time when China's PLAAF (People's Liberation Army Air Force), apart from mentoring the Pakistan Air Force (PAF), is growing exponentially. Besides jointly producing the JF-17 fighter jet with Pakistan, China has in service a fifth generation stealth-capable fighter, the J-20, and is developing the H-20 stealth bomber.

RAPIDLY REARMING IAF

A few months back, the IAF came out with its latest doctrine, which mentions how mere 'strategic depth' failed to stop the IAF's Canberras, launched from Agra, from attacking Pakistan's Peshawar airbase in the 1965 war. Such extended 'reach' can easily be attained now—IAF aircraft such as SU-30 MKI and Rafale can project firepower on to the Malacca and Sunda Straits, while the C-17 and C-130 transport aircraft can reach further afield. The IAF is likely to have around 37-38 fighter squadrons by 2030, with each squadron having 18 aircraft. The target is to get to 42 squadrons by 2038, including six squadrons of the under-development LCA Mk-2 and four of the AMCA. A decision has also been taken to acquire 114 multi-role combat aircraft like the Rafale.

The IAF operates a mix of aircraft fleets (over 1,700 aircraft, including 900-plus combat aircraft) largely of



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Chief, Indian Air Force

foreign origin, ranging from the legacy MiG-21 Bison to Jaguar, Mirage 2000, MiG-29, Su-30MKI, LCA Tejas, and the Rafale. The IAF will start receiving the first of the 83 Tejas Mk-1A from 2024 onwards. With improvements like air to air refuelling, AESA (active electronically scanned array) radar and an EW (electronic warfare) suite, Mk-1A would meet many requirements. Efforts are on to upgrade the Sukhoi Su-30 fleet with the integration of indigenous radar, Infrared Search and Tracker (IRST) and other systems. The fleet of 116 Jaguar fighter-bombers was upgraded last year to enable them to deliver nuclear gravity bombs, making them a part of India's nuclear triad